This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

AR on Ladders: Effect of Performing Augmented Reality Tasks on Postural Sway While Working on a Ladder

Andrew C. Stewart, Wei-An Teng, Joseph L. Gabbard, Michael L. Madigan



Fig 1. Four participants at different points in the study: a baseline balance assessment task (A), a virtual text reading task (B), a virtual button poking task (C), and a block assembly task requiring AR information (D) – all while on a ladder.

Abstract – The use of augmented reality (AR) is growing in industrial occupational settings such as within the construction industry [1, 2]. While AR provides convenience such as providing hands-free visual access to time-sensitive information, it may also introduce unintended adverse consequences. Falls off ladders are already a major source of workplace injuries [3], and any adverse effects of AR on balance while working on a ladder could exacerbate the risk of falling. Therefore, the purpose of this study was to investigate the effects of viewing and interacting with AR on balance during simulated construction tasks while on a ladder. The four tasks investigated involved: (1) a reading task with AR input, (2) an assembly task requiring AR input, (3), an assembly task with verbal input, and (4) a button poking task with virtual AR buttons. We also investigated screen-relative and world-relative AR presentation styles across the relevant AR-support tasks. Balance was assessed by measuring mean speed of postural sway of the center of pressure using a force plate under the ladder, with increases in sway speed from a baseline condition involving standing as still as possible on the ladder being compared between tasks and presentation styles.

Results indicated postural sway was greater during the assembly task with verbal input (6.39 cm/s; $p<0.05$), the assembly task with AR input (6.33 cm/s; $p<0.05$), and the button poking task with AR input (7.35 cm/s; $p<0.05$) as compared to the reading task with AR input (2.90 cm/s). Additionally, postural sway increased more during the button poking task with AR input as compared to the assembly task with AR input. Regarding the two presentation styles, postural sway with a world-relative AR display (9.27 cm/s) was greater as compared to a screen-relative AR display (4.95 cm/s) during the poking task with AR input ($p < 0.05$) but did not differ between presentation styles during the assembly task with AR input ($p > 0.05$) or the reading task with AR input ($p > 0.05$). Our results imply that care must be taken when designing AR applications for applications that are to be used on ladders or other environments where postural stability is of great importance, as the task to be completed and the presentation style can affect a worker's balance. Further research into user interface presentation styles interaction techniques and tasks involving AR menu navigation is proposed.

Index Terms – Augmented reality, mixed reality, extended reality, balance, ladders, postural sway.

1 Introduction

Augmented Reality (AR) is a form of multimodal technology that overlays digital information and graphics overtop a physical environment in real time [3, 4]. This is achieved by one of three methods of projecting a digital environment atop the physical environment: video see-through, optical see-through, and projective displays [4]. These displays are often accompanied by auditory methods of feedback, whether by small speakers or headphones, aiding in their usage [5]. AR interfaces allow for a new paradigm of work that does not require traditional monitors, keyboards, and mice [6].

Within the construction industry, AR has found uses in installation, maintenance, repair, and inspection [7-9]. During these activities it is common to refer to paper instructions, manuals, and plans, which require a worker to hold in-hand or place nearby to access the information. AR enables workers to free both hands and engage with information presented in their environment [9, 10]. Research recommends use of AR for construction tasks because studies have shown that AR can reduce the cognitive load typically associated with remembering or spatially translating time-sensitive information and instructions [10].

Within construction and related occupations, it is common to use ladders to reach workspaces that are overhead or higher than a

- *Andrew Stewart, Wei-An Teng, Joseph Gabbard, and Michael Madigan are with Virginia Tech. E-mail: {andrewcstewart@vt.edu | weianteng3@vt.edu | jgabbard@vt.edu | mimadiga@vt.edu}*

Manuscript received xx xxx. 201x; accepted xx xxx. 201x. Date of Publication xx xxx. 201x; date of current version xx xxx. 201x.
For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org.
Digital Object Identifier: xx.xxxx/TVCG.201x.xxxxxxx/.

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

worker can comfortably reach. Injuries resulting from falls off ladders are common in the workplace, making up over 22,000 workplace injuries and 161 workplace fatalities in 2020 [11]. Within the construction industry alone, over 80% of injuries that required treatment in an emergency room involved falling off a ladder [12]. Therefore, it is reasonable to raise concerns over the possible safety effects of using AR devices while on a ladder. Given that virtual reality tools can lead to balance impairment due to various factors including the vergence-accommodation conflict, a similar effect may occur with AR [13-15]. As such, the use of AR while working on ladders may further increase fall risk.

In this work, we aim to determine to what extent usage of AR while working on a ladder affects postural way, which is an indicator of balance and fall risk. Within the construction industry ladders are essential tools for completing work out of reach of the average worker. Balance impairment has also been reported in optical see through headsets [13, 16]. With an increased presence of AR tools in the workplace, it is essential to understand the safety implications of these devices.

The study presented herein examines four different tasks that aim to simulate potential AR use within the construction industry: a text reading task with AR input, a block assembly task with AR input, the same block assembly task with verbal input, and a button poking task with AR input. The three tasks relying on visual AR input are studied across two different presentation styles: screen-relative and world-relative. A *screen-relative display* remains centered in a user's field of vision regardless of head position while *world-relative displays* remain in a fixed location in space independent of head orientation. Two primary research questions are addressed:

RQ1: What is the effect of simulated AR work tasks on postural sway while working on a ladder?
RQ2: Do the effects of simulated AR work tasks on postural sway differ between screen-relative and world-relative presentation styles?

To the best of our knowledge, there have been no empirical investigations into a user's postural sway while using AR to work on a ladder. Thus, a contribution of this work includes empirical postural sway data associated with performing AR-supported tasks on ladders. Further, we believe that our work can generate discussions on possible negative effects of AR usage in safety-critical situations and open opportunities for follow-up research into ladder fall prevention interventions, tailoring AR tasks to mitigate adverse effects on balance, and the tasks that can safely be performed on a ladder.

2 Background and Related Work

While no studies have compared tasks and presentation styles on postural sway while working on a ladder, similar fields have been explored.

2.1 Postural Sway and Sensory Systems

Balance, while a fundamental function of the body, is a complex process involving mechanical, physiological, anatomical, and psychological components [13]. The visual, vestibular, and proprioceptive sensory systems work together to help maintain upright balance and regain balance if perturbed. Balance is commonly assessed by measuring postural sway, or the natural and unavoidable movements that occur while standing. Postural sway is commonly assessed using a force plate or balance board to track the movement of the center of pressure under the feet while standing [13, 16]. Any changes to the effectiveness of the visual, vestibular, and proprioceptive sensory systems (such as when vision is obscured with AR use), postural sway typically increases. Moreover, the use of mixed reality devices can increase the cognitive load of a user and, in turn, increase postural sway [17, 18].

2.2 Industrial AR Applications

Prior studies suggest AR is promising increased relevance to the construction industry, and as AR technology improves, its usage in the industry (and many others) will increase [19-21]. The increased interest is in part due to the benefit of gesture-limited access to information while using AR to assist in tasks within maintenance and construction [9, 10]. AR can project digital step-by-step instructions directly within a worker's line of sight, which reduces the necessary head movements associated with reading traditional instructions [8]. Importantly, head movements have been linked to causing postural sway through disturbances to the vestibular systems, which are located within the inner ear [22, 23]. Many AR interface require head movements, especially due to current displays' limited field-of-view – thus understanding the impact of AR usage on postural sway can lead to greater insights on the safety of using AR at heights.

Within industrial settings, AR is also finding use in inspection, visualization, and safety training. During building and infrastructure safety inspections, AR can supplement an inspector with deep learning capabilities that allow for real-time damage detection [24]. This framework allows for signs of corrosion and fatigue, like cracks, to be caught automatically and flagged for the inspector. This assistance can lead to greater fault detection and more thorough building and infrastructure inspections, improving public safety.

When the utilities and structures such as columns behind walls and floors are of interest, AR has been used to map out utilities in areas that cannot be viewed easily through a simulated form of "x-ray vision" [3]. Technicians and construction workers can see exactly where pipes, electrical lines, and rebar are located behind walls with this technology, preventing damaging utility lines when working around them [25].

AR has been used to improve workers safety by allowing workers to gain experience dealing with situations that would otherwise be difficult or dangerous to simulate [26]. Workers who experience these scenarios in a first-hand presentation develop a more proactive approach to preventing and managing risks [27]. In respect to ladders, augmented reality has used for ladder safety assessments by analyzing workers' postures in real time [28]. This technology is important for worker safety due to the prevalence of ladder misuse in workplace related injuries [29].

2.3 Occupational Falls Hazards and Ladder Safety

Many falls from ladders are preventable with common causes for falls including incorrect ladder angle, improper ladder selection for the job, insufficient ladder inspection, overreaching, and lack of training [29]. We posit that without care, the use of AR on ladders could result in overreaching, since interacting with AR at very close distances is difficult due physiological issues related to vergence as well as the poor ergonomics associated with bare handed interaction near the body. The CDC notes that novice workers (due to inexperience) and older workers (due to reduced sensory capabilities) are at higher risk of fatal falls. Such risks could be exasperated due to AR usage, especially given that AR shows promise for upskilling novice workers in new work settings [29].

2.4 Physiology of Head-Mounted Displays

When using virtual reality head-worn displays (HWDs), it's been well documented that the mismatch between visual cues of movement from the device and the body's vestibular system not detecting movement is a leading cause of visually induced motion sickness [22, 30]. Many HWDs block or reduce a user's peripheral

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

vision, which is another visual cue that the body uses to maintain balance [13, 31]. Extended use of AR HWDs can result in eye strain and blurred vision, with more dynamic visuals inducing dizziness and ataxia [13, 22, 30]. Currently, some technological limitations can lead to nausea and dizziness, namely latency, lag, and frame rate modulation [22]. Issues of latency and lag are further exacerbated in AR applications, since virtual objects are expected to be accurately registered in the real world.

2.4.1 AR and Postural Sway

Research indicates that HWDs can increase postural sway when compared to viewing a traditional monitor [22, 31]. The postural instability theory suggests that increases in postural sway have strong connections to cybersickness with symptoms being akin to fatigue, hangovers, and influenza and lasting for some time after the headset is removed [32]. From a subjective viewpoint, when using HWDs users report increased fear of falling and being more prone to falling, particularly when virtual objects are not anchored in the real-world [30]. Studies have also pointed to a strong correlation between visually influenced body sway and perceived disorientation [15]. These observations further motivate our project to investigate the usage of AR in scenarios of elevated fall risk.

3 Experimental Overview

The study presented herein uses a 3x2+1 within-subjects design whereby three tasks requiring visual AR information are crossed with two presentation styles. The last condition presents information to participants verbally where presentation style is not relevant (see Section 3.3.2 below for more details). Each in-person session took approximately an hour and a half to complete. The order of presentation of the seven experimental conditions was determined using a balanced Latin square. Each of the seven experimental conditions was performed three times for a total of twenty-one trials. Tasks were performed on a ladder with participants maintaining three points of contact, uniform foot placement on a prespecified rung, and with a rest break after every seven trials. One baseline balance assessment was conducted prior to experimental trials. A force plate beneath the ladder collected center of pressure data over each 30 second trial. Participants used a *Microsoft HoloLens 2* head-worn mixed reality display to assist in the six visual tasks (and kept the display on for the seventh aural task). Participants used standardized sneakers to avoid any effects of differing footwear.

3.1 Participants

A biological sex-balanced convenience sample of 28 participants, aged 18-29 years with a median age of 21 years old, was recruited from the university and local community using flyers posted around campus and email listservs. The biological sex-balancing was achieved through asking participants of their biological sex assigned at birth and gender identity was not collected, thus we use "sex" in this paper. Participant body height and mass were self-reported with median height of 170 cm and median mass of 68kg. 26 participants identified as right-handed, one identified as left-handed, and one identified as ambidextrous, though used their right hand for the trials. Table 1 shows a breakdown of self-reported experience with using ladders and virtual reality devices (median values for both categories being 0-4 uses/year).

Table 1. Participant Experience with Ladders and Virtual Reality

Task Experience	0-4 uses/year	5-14 uses/year	15+ uses/year
Ladder Usage	20	7	1
VR Usage	23	3	3

Participant inclusion criteria included: (1) having no self-reported musculoskeletal injuries within the past 6 months, (2) not heavier than 100 kilograms, (3) not taller than 2 meters, (4) do not wear glasses, though vision corrected with contact lenses was allowed, (5) do not fear heights, and (6) were not pregnant. Participants were excluded if they had recent musculoskeletal injuries or were pregnant to reduce the risk of harm to vulnerable populations. Participants were excluded for height and weight based on safety recommendations of the A-frame ladder being used. Participants were excluded if they wore glasses due to concerns of the HoloLens fitting properly. All participants provided written consent prior to participation, and the study was approved by the Virginia Tech Institutional Review Board.

3.2 Experimental Set-up



Fig 2. Ladder and force plate set-up. The force plate is beneath the metal mounting plate. The baseline balance fixation target can be seen on the wall in the top right corner.

The experimental protocol consisted of two stages: a short task familiarization portion followed by experimental trials on the ladder. During the task familiarization phase, participants were first shown a presentation detailing exactly what they will see when using the HoloLens 2 during the experiment. Once completed, the participant calibrates the HoloLens's eye tracking from instructions delivered by the experimenter. They then stood on marked vertical lines on the ground to simulate proper foot placement while on the ladder, while standing in front of the assembly box. Participants were walked through a practice sequence, performing each of the three tasks once while on the ground to become familiar with the tasks themselves. The results of the practice trials were not analyzed.

Experimental trials were then performed on a ladder on top of a force plate. Two climbing mats were placed next to the ladder in case of a lateral fall. The ladder was secured to metal plate using four bolts at each foot, and the metal plate was bolted directly to the force plate. Participants climbed the ladder up to a designated rung to complete the trials.

3.3 Protocol

3.3.1 Balance Baseline and Task Familiarization

Before any AR use, participants completed a baseline balance trial on the ladder involved holding the top of the ladder with only their

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

dominant hand with their nondominant hand at their side. Participants were instructed to stand as still as possible, without talking, while staring at a fixation target recommended for stability testing [33]. Each baseline collected 30 seconds of balance data at 500 scans/second. Once completed, participants are instructed to descend the ladder.

Participants were asked to don and adjust the HoloLens until it was snug on their head. Participants were then instructed to look at their hands through the device to ensure their hands are being properly tracked before the experiment begins. The experimenter walked participants through the eye calibration process and had them raise the visor after completion. At this point, a short presentation detailing the exact screens that are displayed in the experiment were shown to participants, so they knew exactly what to expect when the experiment began. After the familiarization view, participants walked around to a taped "H" on the ground, positioned their feet in the same way they will need to later the ladder, and began the interactive phase of task familiarization. This phase began with a short calibration (that established the world relative frame of reference for each participant based on head position) and then presented each task and presentation style once so participants understood what actions were needed to perform for each trial. Once this phase was complete, participants raised the visor and was ready to commence to the formal experimental trail phase.

3.3.2 Experimental Trials

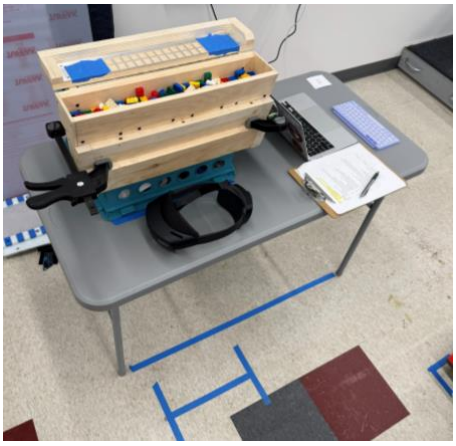


Fig 3. Assembly box in the task familiarization set-up.

Researchers first removed the assembly box (Figure 3) from the task familiarization table and clamped it to the top of the ladder. The force plate beneath the ladder was then calibrated with the box attached and zeroed out after which each participant climbed the ladder to the rung marked with tape. Participants then lowered the visor, and the AR testbed application was launched remotely from the experimenter's laptop. Participants then performed a second calibration to relocate the world-relative frame of reference based on the participants' new head position atop the ladder. Once the experimenter verbally confirmed that the participant was ready, the AR testbed application began the task sequence.

As previously mentioned, participants completed seven experimental conditions: a reading, block assembly and button poking task presented separately as either screen-relative or world-relative graphics, as well as an additional block assembly task where assembly order was conveyed verbally by the experimenter (Table 2).

Table 2. Seven experimental conditions to be completed on the ladder.

Tasks	Input
Text Reading	**Visual via AR, screen-relative**
Text Reading	**Visual via AR, world-relative**
Poking Virtual Buttons	**Visual via AR, screen-relative**
Poking Virtual Buttons	**Visual via AR, world-relative**
Assembling Physical Blocks	**Visual via AR, screen-relative**
Assembling Physical Blocks	**Visual via AR, world-relative**
Assembling Physical Blocks	**Verbal input from Experimenter**

These seven experimental conditions were chosen to simulate different types of tasks that could be completed on a ladder with differing degrees of user demand. The task that required the least integration with the AR was text reading, meant to simulate reading instructions or a manual. A task with increasing user demand was the button poking task, meant to simulate AR menu navigation. The task with the most user demand was the block sorting task, as it required both referring to displayed information and performing a fine-motor task. The block assembly with verbal input simulated a partner instructed task and was designed to see if a block assembly task with verbal input would be more stable than with AR input.

Each task included a five second countdown and the task would be presented for 40 seconds. This amount of time was chosen to allow for at least six seconds of "ramp-up" time and 30 seconds of usable data, with a four second buffer. After the 40 second trial, the AR application automatically ends the task. All tasks were designed to take longer than 40 seconds to prevent participants from finishing tasks early and standing idly. Participants were time-pressured to complete tasks as accurately as possible but were not expected to complete the tasks.

All AR interfaces were presented ~55cm from the user's head with the world-relative presentation style anchoring based on calibration phase. Once anchored, the world-relative interfaces did not move. This distance was determined during pilot testing and aimed to balance participants' ability to reach the button poking interface comfortably while also presenting other task interfaces at comfortable distances for both reading and assisting in the assembly task.

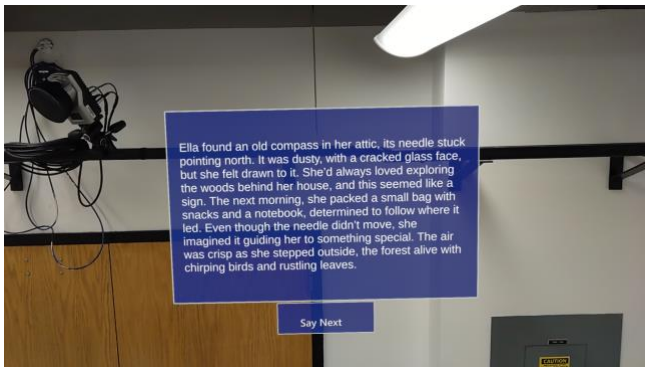


Fig 4. Participant's point of view of the text reading task.

Text Reading Task. During the reading tasks, participants were instructed to speak clearly and loudly, as if they were giving a speech or a presentation. A paragraph was displayed via HoloLens2 which participants read aloud. Three paragraphs containing four to six sentences per trial were generated with OpenAI's *ChatGPT* using a prompt for a warm, fantasy story in English. Once they reached the end, participants uttered the phrase "Next" to cue the experimenter to manually administer the next prompt, as we did not want to rely on the HoloLens's speech recognition to advance tasks.

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

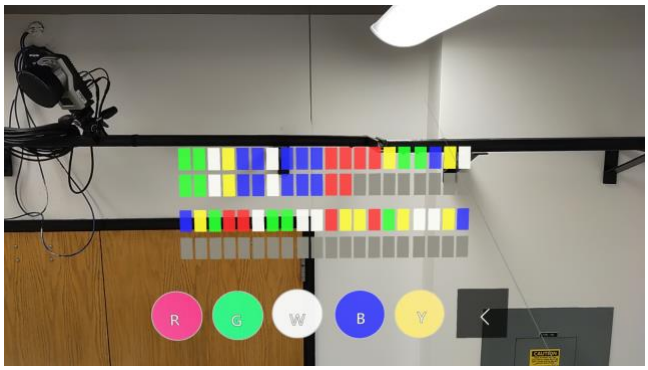


Fig 5. Participant's point of view of the button poking task.

Button Poking Task. During the interactive button poking task, participants used their dominant hand to match a color sequence as presented via AR by poking corresponding virtual buttons located beneath the sequence. The sequences included red, yellow, green, blue, and white squares, and the buttons available reflected these five colors. Participants were also given a backspace button and told that use of the backspace was encouraged in the event of a mistake. As with the previous tasks, if participants completed the task before the forty second period was complete, they said "Finish."



Fig 6. Participant's point of view of the block assembly task.

Block Assembly Task. During the block assembly tasks, participants use their dominant hand to place the blocks in the sorting platform in the exact same order shown via the AR display. The block sequences displayed included red, yellow, green, and blue blocks in randomized orders. The physical blocks available to participants included the same four colors, as well as white blocks to act as a hinderance. In the case of the participant completing the task before the forty second period is complete, they were instructed to say "Finish." The sorting platform is then dumped back into the block bin in preparation for future trials.

The block sorting task that used verbal instructions followed the same protocol as above, but instead of the participant using AR visual information, the experimenter read a color sequence aloud – reading one color at a time as participants assembled blocks. The next color was spoken when the experimenter saw the previous block placed in a location. In this condition participants wore the HoloLens display but there was no visual information required for the task nor displayed via the display.

Upon completion of trial twenty-one, participants descended the ladder, returned the shoes, and were compensated for their time.

3.4 Dependent Measures

Two dependent variable was used to evaluate the effect of different tasks—postural sway and accuracy. Postural sway was determined as the mean speed of the center of pressure measured using the force plate under the ladder. The center of pressure (COP) is the location on the surface of the force plate that represents the mean position of forces applied. As a participant stands and sways on the ladder, the COP on the force plate moves. The mean speed of the COP during trials is a common and sensitive measurement of postural sway and fall risk [16, 34, 35]. The center of pressure mean speed data was collected with a force plate [Model 4550-08 Bertec, Corp. Columbus, OH, USA] beneath the ladder set-up. The voltages collected from the force plate were amplified [Model AM-6100, Bertec Corp. Columbus, OH, USA] and sampled at 500 Hz using data acquisition system [National Instruments, Austin, TX, USA] for post-processing. COP mean speed was determined using customized MatLab™ code. To isolate the effects of the tasks on postural sway and to account for individual differences, the mean COP speed from each participant's baseline balance trial was subtracted from the mean COP speed of each experimental trial.

Accuracy was assessed three different ways corresponding to each task. For the reading task, a recording was taken of the participant speaking each passage aloud. This recording was compared to the passage being read and measured accuracy by dividing the number of correct words by the total number of words spoken, as not to inadvertently penalize participants on lower completion rates. For the block sorting tasks, a photo was taken of the sorting platform after each trial and compared to an output file generated by the testing application listing the sequence of colors displayed to the participant. This accuracy was determined by the number of correct blocks placed divided by the total number of blocks placed. For the button poking task, the testing application produced the sequence inputted by the participant alongside the sequence displayed for direct comparison. Accuracy was calculated by dividing the number of correct colors selected by the total number of colors of the sequence completed. Backspaces were not counted against the participant's accuracy score.

4 Results

Four participants were excluded from analysis due to equipment malfunction or participants not following instructions. Statistical analysis was conducted using *JMP*. A mixed-effects model with REML estimation was used to analyze the effects of presentation style, task, and their interaction on participants' increase in postural sway from their baselines. Nominal data included: Sex, Experience using ladders, Experience with VR, Trial number, Task, Presentation Style, and Repetition. Continuous data included: COP mean velocity subtracted from baseline and accuracy. Data that did not pass tests for normality was transformed with a square root transform.

The fixed effect tests showed that the two measures of prior experience, experience using ladders ($F(1,21)=0.63$, $p=0.438$) and experience using VR ($F(2,21,)=0.88$, $p=0.431$) did not have an effect on postural sway. Trial number also did not have an effect on postural sway ($F(20,398.2)=1.48$ $p=0.086$), suggesting that no systematic order effects due to, for example, learning or fatigue were seen across trials. Additionally, task accuracy ($F(1,406.6)=0.005$, $p=.944$) did not have a significant effect on postural sway. The following results focus on difference in COP mean velocity from baseline as the dependent measure.

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

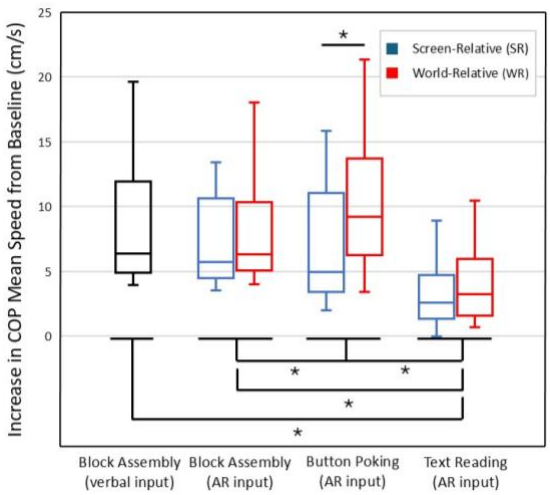


Fig 7. Increase in postural sway from baseline across tasks and presentation styles. Brackets above the box and whisker plots indicate differences between presentation styles within each task, and brackets below the box and whiskers plots indicate differences between tasks (inclusive of both presentation styles). All asterisks over brackets indicate statistical significance at $p < 0.05$. Error bars show 95% confidence intervals.

Figure 7 uses box and whisker plots to illustrate postural sway median, upper and lower quartiles, and range of each of four tasks.

4.1 Main Effects of Presentation Style

There was a main effect of presentation style on COP mean speed, $F(1,398)=58.02$, $p<0.0001$. Pairwise comparisons using Tukey HSD indicated the world-relative presentation style increased postural sway (an adverse effect on balance) more than the screen-relative presentation style, with the difference demonstrating statistical significance. Block assembly with verbal input was excluded from this analysis due to not having either world-relative or screen-relative presentation styles.

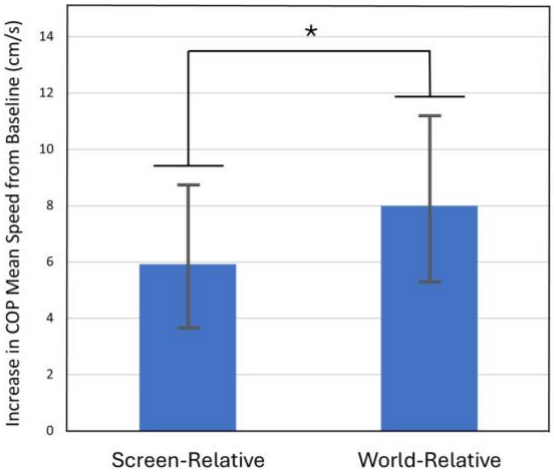


Fig 8. Increase in postural sway from baseline across presentation styles. All asterisks over brackets indicate statistical significance at $p < 0.05$. Error bars show 95% confidence intervals.

4.2 Main Effects of Task

A main effect of task was observed, $F(2,398)=172.49$, $p<0.0001$. Pairwise comparisons using Tukey HSD indicated the task being performed has a measurable effect on a participant's postural sway. To conduct this comparison with the block assembly with verbal input, world-relative and screen-relative presentation styles were combined. The button poking task with AR input demonstrated the greatest increase in postural sway, followed by block assembly with verbal input, block assembly with AR input, and text reading with AR input. The differences between the test reading task and all other tasks were statistically significant, as well as the difference between block assembly with AR input and button poking with AR input.

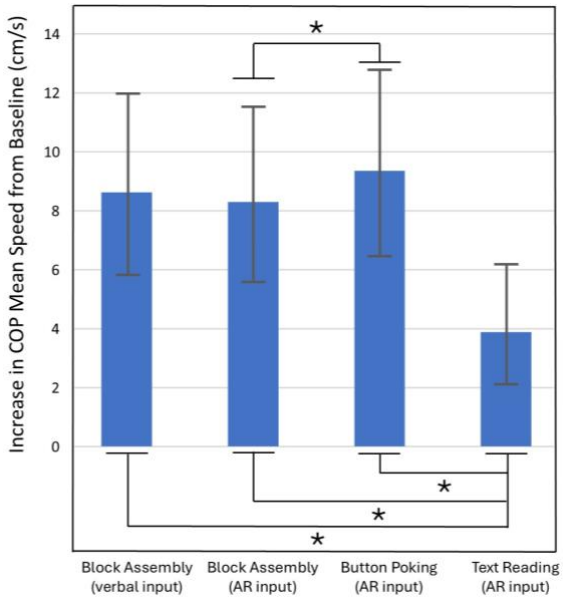


Fig 9. Increase in postural sway from baseline across tasks. All asterisks over brackets indicate statistical significance at $p < 0.05$. Error bars show 95% confidence intervals.

4.3 Interaction Effects

The main effects are superseded by a significant Task × Presentation Style interaction $F(20,398)=14.28$, $p<0.0001$ indicating the effect of presentation style on COP mean speed differed across tasks. Across both world-relative and screen relative presentation styles, the button poking task with AR input demonstrated the greatest postural sway deviation from the baseline when compared to the text reading with AR input and both forms of the assembly tasks. In particular, the button poking task presented in a world-relative presentation style showed significantly higher sway than all other task and presentation style combinations. Table 3 illustrates the differences in COP mean speeds, and the lower and upper confidence limits of the difference. Block assembly with verbal input was excluded due to not having a world-relative or screen-relative presentation style.

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

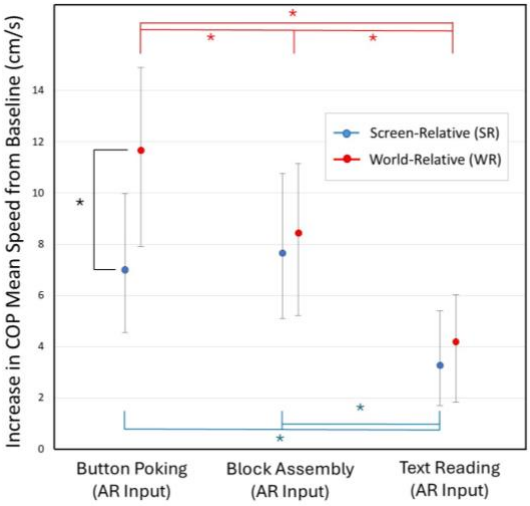


Fig 10. Increase in postural sway from baseline across tasks and presentation styles. Brackets are color-coded for each presentation style.

Table 3. Least squares means differences between world-relative button poking and all other task and presentation style combinations.

Task/Presentation Style	Δ mean	Δ Lower CL	Δ Upper CL
Button Poking (AR)/WR	-	-	-
Button Poking (AR)/SR	**0.776**	**0.521**	**1.030**
Block Assembly (AR)/WR	**0.488**	**0.233**	**0.743**
Block Assembly (AR)/SR	**0.634**	**0.380**	**0.888**
Text Reading (AR)/WR	**1.350**	**1.096**	**1.605**
Text Reading (AR)/SR	**1.605**	**1.349**	**1.861**

Text reading tasks had the lowest postural sway overall; both screen-relative and world-relative presentation styles differed significantly from all button poking and block assembly tasks. The two text reading tasks were also not statistically significant from one another (Δ mean = 0.255, Δ std err = 0.090, 95% CI [-0.002, 0.511]), suggesting presentation style had little effect on postural sway when reading text. In contrast, the button poking task in a world-relative presentation style showed significantly greater postural sway than in the screen-relative presentation style, suggesting that the world-relative presentation style was disorienting when participants were engaging with buttons in the AR space. The assembly task with AR input also presented this trend of the world-relative presentation style being less stable than the screen-relative presentation style, though the difference between presentation styles was not statistically significant.

4.4 Other Findings

Secondary analyses included sex as an independent variable, which revealed significant Sex × Task ($F(2,413)=13.91$, $p<0.0001$) and Sex × Presentation Style ($F(1,413)=5.78$, $p=0.017$) interactions despite no significant main effect of sex on COP mean speed ($p = 0.097$), meaning that detectable difference in COP is due to differences in COP across tasks and presentation styles, not sex.

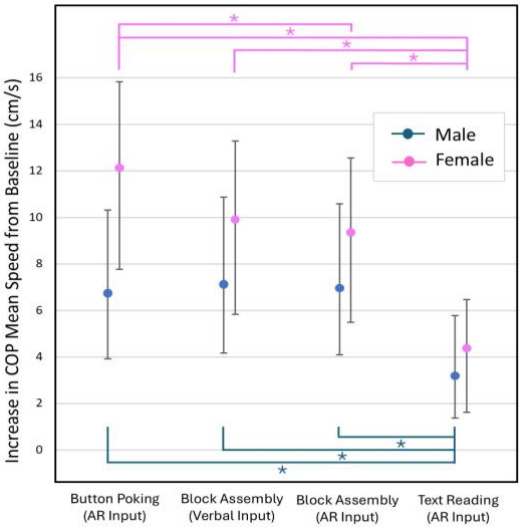


Fig 11.Increase in postural sway from baseline across tasks between male and female participants. Brackets are color-coded to differentiate males and females.

When comparing performance across tasks and sex, we observed significant differences in COP mean speed between text reading (AR input) and block assembly (both verbal and AR input), as well as text reading (AR input) and button poking (AR input). However, there was a significant difference between block assembly (AR input) and button poking (AR input) for female participants that was not seen in male participants.

With regards to presentation style, trends suggest female participants saw a larger increase in postural sway from their baseline under the world-relative presentation style than the screen-relative presentation style (LS Means difference = 0.50) when compared to male participants (LS Means difference = 0.26). This suggests that presentation style may affect postural stability differently dependent on sex, though a thorough analysis of these effects is outside the scope of this study.

5 Discussion

The purpose of this study was to investigate the effects of AR on balance during simulated construction tasks while on a ladder. We explore how we accomplished this in reference to our research questions.

5.1 RQ1: What is the effect of simulated AR work tasks on postural sway while working on a ladder?

To address this question, we look back at the data we collected and the main effects of task on postural sway. Task had the strongest effect in the entire model ($F(2, 398)=172.49$, $p<0.0001$), suggesting that the physical and/or AR-based tasks that a worker is doing while on a ladder play a key role in their postural stability. The button poking tasks saw the greatest increase in postural sway from baseline, suggesting that interacting with AR interfaces (that require virtual button presses) may cause instability, and at least is associated with more instability than reading AR text (and more instability than the block assembly task for world-relative condition). The block assembly task saw moderate sway, and the text reading task saw the least postural sway increase. Providing visual instruction visually via AR did not result in any measurable differences in postural sway as compared to providing oral instructions.

To understand this behavior, two explanations are considered: reaching and cognitive load. Leaning while using a ladder is a

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

known safety concern for falls, and our button poking task required the greatest reach forward of three tasks. The literature suggests that participants are more likely to physically maneuver their body's to a posture that allows for the easiest interactions with the AR interface [29, 36, 37], thus we expected that users were more likely to lean forward to poke the AR interface, and when doing so, may introduce more lateral instability due to the fact that A-frame ladders are prone to more lateral instability than longitudinal. While the block assembly required more hand movements, and constant checks to visual instructions, the hand movements did not require reaching, and in fact were limited to the assembly box space located directly atop the ladder. Note that this location is closer to participants' (and ladder) center of gravity. If the assembly task had been placed in front as opposed to atop of the ladder (for example, at the same distance as the poking task), different results may have been observed.

Further, we believe that the use of the assembly box may have given a visual and physical anchor fixed to the ladder for participants to retain balance, whereas the AR interface used for the poking task had no similar affordances. Research indicates that cognition is a component in postural control, and when mental facilities are engaged, postural sway increases with less bandwidth to be applied to proprioceptive responses [38]. The "posture first" principle dictates that individuals will prioritize balance over a cognitive task, though this can be transgressed if a participant has a sense of safety, such as maintaining three points of contact or being in a laboratory setting [38, 39].

The results we collected and the justifications we have found imply that tasks that involve interacting with an AR interface can lead to greater instability as compared to simply standing on a ladder. While this should be considered in the development of AR applications for use in the construction industry—as more interaction with an AR interface a worker does while on a ladder can lead to greater postural sway and thus risk of imbalance—it is important to frame the increase in postural sway in the performance of interaction-heavy tasks. While heavy interaction with an AR interface implies an increase in postural stability, this may not be generalizable or applicable for all AR designs.

5.2 RQ2: Do the effects of simulated AR work tasks on postural sway differ between screen-relative and world-relative presentation styles?

From our data, we saw a highly significant effect of presentation style suggesting that how information is presented in space can have a meaningful influence on a worker's postural sway while on a ladder. Most of these effects are attributable to button poking task where the world-relative presentation style produced higher postural sway than screen-relative, implying that when an interactive heavy AR interface stays fixed within the environment, participants displayed greater postural sway.

When analyzing these results, we explored the motion participants exhibited on the ladder. When tasks were presented in a screen-relative presentation style, users kept their heads very stable to clearly see the AR projection. In order to focus on colors and text, the eyes rely on foveal vision, located directly in the center of our field of vision [40, 41]. When the head must be kept stable, we believe viewing the AR input requires the user's eyes to scan without moving their head. In a world-relative presentation style, the user is free to move their head while parsing visual information and can even move their head and shoulders enough to where the motions required to poke the interface become more exaggerated than those associated with screen-fixed buttons. For example, if users lean back a little while atop the ladder using the world-relative condition they suddenly will need to reach a little father. Further, studies have demonstrated that when information is presented in a format wider than our direct parafoveal vision, head movement is significantly increased [42]. This kind of head movement can lead to greater postural sway from perturbations to the vestibular systems [13, 43].

Looking beyond the implications for safety on ladders, our findings also contribute to the broader discussion of screen-relative versus world-relative AR user interface design. Previous work has discussed tradeoffs between these two styles with world-relative user interfaces often favored when designers wish to prioritize increased spatial contextual understanding [44], reduced cognitive tunneling and better directed attention toward objects in the environment [45]. Conversely, screen-relative interfaces are often preferred when prioritizing the need to reacquire user interface elements just after head movements and when efficient interaction is desired [46, 47]. Our results suggest that the *physical stability demands* imposed by each presentation style are an additional design tradeoff consideration. While world-relative AR user interfaces offer advantages for maintaining spatial context, they may (depending on the user interface design) encourage larger head, shoulder, and upper-body movements during interactive tasks, which in situations requiring balance (e.g., ladder work) could increase postural sway. In contrast, screen-relative interfaces may encourage users to stabilize their head while interacting with an AR user interface, potentially reducing upper body motion during poking interactions. These findings suggest that design consideration between screen-relative and world-relative presentation styles should not depend solely on perceptual or cognitive considerations, but also on the physical demands and safety requirements of the work environment.

While the difference between postural sway in world-relative and screen-relative presentation styles were similar for the block assembly and text reading tasks, they varied significantly in the button poking task, as seen by the significant interaction effect. This interaction effect implies that care must be taken in the design of AR applications for use in the construction industry where ladder is common. Applications requiring poking should consider screen-relative displays that do not obstruct a worker's vision but maintain their position relative to the worker's head orientation. World-relative presentation styles should be reserved for AR interactions on ladders that do not require reaching, or for construction tasks that not performed on ladders [20, 21].

5.3 Other Findings

Considering the findings of Sex × Task and Sex × Presentation Style interaction effects, these patterns suggested that any differences in COP mean speed between males and females was due to COP mean speed differences between tasks and presentation styles. Nonetheless, the trends observed between sexes may have contributed to the interaction effect between Task × Presentation Style. Further research into the differing effects of tasks and presentation styles on postural sway between sexes may reveal trends not seen in this study.

6 Limitations and Future Work

Some of the limitations of this study include the sample demographics not aligning with demographics of construction workers who would be using this AR technology in the field. The demographics of workers at a higher risk of ladder falls includes older Hispanic males and self-employed workers [48]. A wider range of participant ages would generate more generalizable results. The study was conducted in a laboratory setting, and therefore did not include typical workplace environmental factors, such as noise, temperature, humidity, and other distractors. The ladder assembly, while useful to simulate an assembly task at the top of a ladder, does not simulate a work task such as installation or wiring that involves working against a surface next to the ladder or overhead. The study also did not collect subjective balance data,

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

such as perceived instability or symptoms of cybersickness, which could provide useful insights into confidence and perceptions of safety. Measures such as dizziness or cognitive load via a NASA-TLX could provide insight into the cognitive strain placed on participants.

While all tasks were presented directly in front of the participant, this obscures the user's view. A follow-up involving world-relative presentation could place the display at different angles around participants, encouraging them to rotate their head greater angles to view information. This study focused on interacting with AR interfaces using gesture controls like poking. A further exploratory study could explore the difference in postural sway between a task completed with gesture controls as opposed to voice controls. The implications of such further research can help advise AR application design for worker safety.

7 Conclusion

While augmented reality becomes a tool for the construction industry, its effects on postural sway while using a ladder is understudied. In this study we designed four tasks to be completed while on a ladder and measured the change in participant's postural sway across tasks and presentation styles from a baseline. We used an A-frame ladder on top of a force plate to measure postural sway. We found significant main effects of presentation style and of task on postural sway as well as an interaction effect. Difference in postural sway was greatest during a button poking task in a world-relative presentation style. Our results aim to shed light on the potential risks of AR usage while on a ladder, as well as guide AR application development for the construction industry with safety standards in mind.

Acknowledgments

This work was supported by the efforts of undergraduate research assistants Alex Castro, Gigi Genco, Arjun Tuli, and Kevin Wei, and software developed by Wei-An Teng.

References

[1] J. M. Davila Delgado, L. Oyedele, T. Beach, and P. Demian, "Augmented and virtual reality in construction: Drivers and limitations for industry adoption," *Journal of construction engineering and management,* vol. 146, no. 7, p. 04020079, 2020.

[2] F. De Pace, F. Manuri, and A. Sanna, "Augmented reality in industry 4.0," *Am. J. Comput. Sci. Inf. Technol,* vol. 6, no. 1, p. 17, 2018.

[3] R. T. Azuma, "A survey of augmented reality," *Presence: teleoperators & virtual environments,* vol. 6, no. 4, pp. 355–385, 1997.

[4] D. Van Krevelen and R. Poelman, "A survey of augmented reality technologies, applications and limitations," *International journal of virtual reality,* vol. 9, no. 2, pp. 1–20, 2010.

[5] S. Oviatt, "User-centered modeling and evaluation of multimodal interfaces," *Proceedings of the IEEE,* vol. 91, no. 9, pp. 1457–1468, 2003.

[6] G. Yenduri *et al.*, "Spatial Computing: concept, applications, challenges and future directions," *arXiv preprint arXiv:2402.07912,* 2024.

[7] A. Heinzel, S. Azhar, and A. Nadeem, "Uses of augmented reality technology during construction phase," in *Conference: 9th International Conference of Construction in the Twenty-first Century (CITC9), Dubai, United Arab Emirates*, 2017, pp. 1–8.

[8] S. Henderson and S. Feiner, "Exploring the benefits of augmented reality documentation for maintenance and repair," *IEEE transactions on visualization and computer graphics,* vol. 17, no. 10, pp. 1355–1368, 2010.

[9] A. Malta, T. Farinha, and M. Mendes, "Augmented reality in maintenance—history and perspectives," *Journal of Imaging,* vol. 9, no. 7, p. 142, 2023.

[10] A. Seeliger, L. Cheng, and T. Netland, "Augmented reality for industrial quality inspection: An experiment assessing task performance and human factors," *Computers in Industry,* vol. 151, p. 103985, 2023.

[11] B. o. L. Statistics. "Fatal injuries from ladders down in 2020; nonfatal ladder injuries were essentially unchanged." U.S. Department of Labor. https://www.bls.gov/opub/ted/2022/fatal-injuries-from-ladders-down-in-2020-nonfatal-ladder-injuries-were-essentially-unchanged.htm (accessed March 4, 2026).

[12] H. J. Lipscomb, A. L. Schoenfisch, and K. S. Shishlov, "Non-fatal contact injuries among workers in the construction industry treated in US emergency departments, 1998-2005," *Journal of safety research,* vol. 41, no. 3, pp. 191–195, 2010.

[13] P. Epure *et al.*, "Effect of the Oculus Rift head mounted display on postural stability," in *The 10th International Conference on Disability Virtual Reality & Associated Technologies: Proceedings*, 2014: Reading University Press, pp. 119–127.

[14] C. L. Hughes, C. Fidopiastis, K. M. Stanney, P. S. Bailey, and E. Ruiz, "The psychometrics of cybersickness in augmented reality," *Frontiers in Virtual Reality,* vol. 1, p. 602954, 2020.

[15] B. D. Lawson and K. M. Stanney, "Editorial: Cybersickness in Virtual Reality and Augmented Reality," (in English), *Frontiers in Virtual Reality,* Editorial vol. Volume 2 - 2021, 2021–October–13 2021, doi: 10.3389/frvir.2021.759682.

[16] E. Chiarovano, C. De Waele, H. G. MacDougall, S. J. Rogers, A. M. Burgess, and I. S. Curthoys, "Maintaining balance when looking at a virtual reality three-dimensional display of a field of moving dots or at a virtual reality scene," *Frontiers in neurology,* vol. 6, p. 164, 2015.

[17] S. Blomqvist, S. Seipel, and M. Engström, "Using augmented reality technology for balance training in the older adults: a feasibility pilot study," *BMC geriatrics,* vol. 21, no. 1, p. 144, 2021.

[18] A. Z. Kolaei, E. Hedayati, M. Khanzadi, and G. G. Amiri, "Challenges and opportunities of augmented reality during the construction phase," *Automation in Construction,* vol. 143, p. 104586, 2022.

[19] S. Rankohi and L. Waugh, "Review and analysis of augmented reality literature for construction industry," *Visualization in Engineering,* vol. 1, no. 1, p. 9, 2013.

[20] S. Ahmed, "A review on using opportunities of augmented reality and virtual reality in construction project management," *Organization, technology & management in construction: an international journal,* vol. 10, no. 1, pp. 1839–1852, 2018.

[21] H. Nassereddine, A. S. Hanna, D. Veeramani, and W. Lotfallah, "Augmented reality in the construction industry: use-cases, benefits, obstacles, and future trends," *Frontiers in Built Environment,* vol. 8, p. 730094, 2022.

[22] V. M. Schroeder and L. A. T. Biggs, "Fatigue and Eye Strain Associated with Virtual Reality and Augmented Reality Systems," *Adapting Virtual Reality and Augmented Reality Systems for Naval Aviation Training,* vol. 4, p. 104, 2018.

[23] J. Allum, M. Gresty, E. Keshner, and C. Shupert, "The control of head movements during human balance corrections," *Journal of Vestibular Research,* vol. 7, no. 2-3, pp. 189–218, 1997.

[24] S. Wang, S. A. Zargar, and F.-G. Yuan, "Augmented reality for enhanced visual inspection through knowledge-based deep learning," *Structural Health Monitoring,* vol. 20, no. 1, pp. 426–442, 2021.

[25] J. Xu and F. Moreu, "A review of augmented reality applications in civil infrastructure during the 4th industrial revolution," *Frontiers in Built Environment,* vol. 7, p. 640732, 2021.

[26] R. E. Pereira, M. Gheisari, and B. Esmaeili, "Using panoramic augmented reality to develop a virtual safety training environment," in *Construction research congress 2018*, 2018, pp. 29–39.

[27] S. Bhoir and B. Esmaeili, "State-of-the-art review of virtual reality environment applications in construction safety," *AEI 2015,* pp. 457–468, 2015.

[28] H. Iyer, J. Isingizwe, R. Eiris, and H. Jeong, "Ladder safety assessment using head-mounted 360-degree camera-based posture estimation overlayed real-time in augmented reality," in *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 2024: IEEE, pp. 1–4.

[29] N. I. f. O. S. a. Health. "Ladder Safety." Centers for Disease Control and Prevention. https://www.cdc.gov/niosh/falls/ladder/index.html (accessed.

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Visualization and Computer Graphics. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TVCG.2026.3747528

[30] M. Kaufeld, M. Mundt, S. Forst, and H. Hecht, "Optical see-through augmented reality can induce severe motion sickness," *Displays,* vol. 74, p. 102283, 2022.

[31] S. M. Shahnewaz Ferdous, T. I. Chowdhury, I. M. Arafat, and J. Quarles, "Static rest frame to improve postural stability in virtual and augmented reality," *Frontiers in Virtual Reality,* vol. 1, p. 582169, 2021.

[32] S. Davis, K. Nesbitt, and E. Nalivaiko, "A Systematic Review of Cybersickness," presented at the Proceedings of the 2014 Conference on Interactive Entertainment, Newcastle, NSW, Australia, 2014. [Online]. Available: https://doi.org/10.1145/2677758.2677780.

[33] L. Thaler, A. C. Schütz, M. A. Goodale, and K. R. Gegenfurtner, "What is the best fixation target? The effect of target shape on stability of fixational eye movements," *Vision Research,* vol. 76, pp. 31–42, 2013/01/14/ 2013, doi: https://doi.org/10.1016/j.visres.2012.10.012.

[34] F. Quijoux *et al.*, "A review of center of pressure (COP) variables to quantify standing balance in elderly people: Algorithms and open-access code," *Physiological reports,* vol. 9, no. 22, p. e15067, 2021.

[35] F. Quijoux *et al.*, "Center of pressure displacement characteristics differentiate fall risk in older people: A systematic review with meta-analysis," *Ageing Research Reviews,* vol. 62, p. 101117, 2020.

[36] C. L. Deschler, E. M. Pliner, D. L. Sturnieks, S. R. Lord, and K. E. Beschorner, "Correlations between reach, lean and ladder tipping risk," *Journal of biomechanics,* vol. 150, p. 111508, 2023.

[37] R. A. Partridge, A. S. Virk, and R. E. Antosia, "Causes and patterns of injury from ladder falls," *Academic emergency medicine,* vol. 5, no. 1, pp. 31–34, 1998. [Online]. Available: https://onlinelibrary.wiley.com/doi/pdfdirect/10.1111/j.1553-2712.1998.tb02571.x?download=true.

[38] J. Barra, A. Bray, V. Sahni, J. F. Golding, and M. A. Gresty, "Increasing cognitive load with increasing balance challenge: recipe for catastrophe," *Experimental brain research,* vol. 174, no. 4, pp. 734–745, 2006.

[39] G. Andersson, J. Hagman, R. Talianzadeh, A. Svedberg, and H. C. Larsen, "Effect of cognitive load on postural control," *Brain research bulletin,* vol. 58, no. 1, pp. 135–139, 2002.

[40] E. E. Stewart, M. Valsecchi, and A. C. Schütz, "A review of interactions between peripheral and foveal vision," *Journal of vision,* vol. 20, no. 12, pp. 2–2, 2020.

[41] M. Lavidor and V. Walsh, "The nature of foveal representation," *Nature Reviews Neuroscience,* vol. 5, no. 9, pp. 729–735, 2004.

[42] F. A. Proudlock, H. Shekhar, and I. Gottlob, "Coordination of eye and head movements during reading," *Investigative ophthalmology & visual science,* vol. 44, no. 7, pp. 2991–2998, 2003.

[43] C.-J. Chang, T.-F. Yang, S.-W. Yang, and J.-S. Chern, "Cortical modulation of motor control biofeedback among the elderly with high fall risk during a posture perturbation task with augmented reality," *Frontiers in aging neuroscience,* vol. 8, p. 80, 2016.

[44] M. Billinghurst, J. Bowskill, N. Dyer, and J. Morphett, "Spatial information displays on a wearable computer," *IEEE Computer Graphics and Applications,* vol. 18, no. 6, pp. 24–31, 2002.

[45] M. Yeh, C. D. Wickens, and F. J. Seagull, "Target cuing in visual search: the effects of conformality and display location on the allocation of visual attention," (in eng), *Hum Factors,* vol. 41, no. 4, pp. 524–42, Dec 1999, doi: 10.1518/001872099779656752.

[46] T. Höllerer *et al.*, "User interface management techniques for collaborative mobile augmented reality," *Computers & Graphics,* vol. 25, no. 5, pp. 799–810, 2001/10/01/ 2001, doi: https://doi.org/10.1016/S0097-8493(01)00122-4.

[47] J. Peereboom, W. Tabone, D. Dodou, and J. de Winter, "Head-locked, world-locked, or conformal diminished-reality? An examination of different AR solutions for pedestrian safety in occluded scenarios," *Virtual Reality,* vol. 28, no. 2, p. 119, 2024/05/27 2024, doi: 10.1007/s10055-024-01017-9.

[48] C. K. Chaumont Menéndez, J. W. Collins, P. Simeonov, and C. M. Socias-Morales, "Occupational Ladder Fall Injuries–United States, 2011," 2014.

Authorized licensed use limited to: to IEEExplore provided by University Libraries | Virginia Tech. Downloaded on October 30,2026 at 01:16:35 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted,
but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.